Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think>

- The patient has community-acquired pneumonia, which means the infection is from a non-invasive source in the community, not from a healthcare setting.
- Cephalexin, an antibiotic, is used for treatment, indicating a bacterial cause rather than viral or fungal.
- The pathogen elicits a TLR5 activation in macrophages. TLR5 recognizes peptidoglycan, which is a component of bacterial cell walls.
- Since TLR5 is specifically mentioned, this is a key clue to consider peptidoglycan as the major antigen.
- The immune response is described as activated rather than evaded, and TLR5 activation is more specific to bacteria than evasive mechanisms like encapsulation, which would be associated with viruses or some bacteria.
- The response does not specifically mention Th1, but the context suggests a broader immune response rather than just Th1-specific.
- Therefore, the best description is that the pathogen activates TLR5 on the surface of macrophages.

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The best option is C. Peptidoglycan is its major antigen that induces an immune response.

</think>

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